Answers and talk notes

Determine the Volume of Rectangular Prisms

What your student is learning

Tonight we work the volume formula backwards to find a missing edge — and we keep fractional edges exact instead of rounding them away.

Answers

Ask your student to explain before you look.

1. **C.** Multiply 5 × 3. Multiply 5 × 3 to get the base area, then divide the volume by it.
2. **A.** 2 cm. 2 cm: the base area is 4 × 3 = 12, and 24 ÷ 12 = 2.
3. **B.** 4 in.. 4 in.: 72 ÷ 18 = 4.
4. **D.** 9 cubic ft. 9 cubic ft: 2 × 3 = 6 and 6 × 1.5 = 9.
5. **B.** 3 cm. 3 cm: 5 × 3 = 15 and 45 ÷ 15 = 3.
6. **A.** 4 cubic in.. 4 cubic in.: exact is 4 × 2 × 2.5 = 20, rounded is 4 × 2 × 2 = 16, and 20 − 16 = 4.

Talk about it

How do you find a missing edge when you know the volume and two other edges?

Home mission

Pick a small box. Pretend its volume is 100 cubic centimeters and two edges are 5 cm and 4 cm. Can your student find the third edge, then check by multiplying all three?

When they are stuck, ask

- How many edges does the formula use? Did you use all of them?
- What do the two edges you know multiply to?
- Did you round a fraction? Put the exact value back in.

Try not to say

- Add the three edges to get the volume.
- Round the half foot, it will not matter.
- Two edges is close enough for a box.